

Table 3. Bubble departure diameters at 50 kW m⁻² and 100 kW m⁻² including standard deviation.

Sample	D_b @ 50 kW m ⁻² (mm)	D_b @ 100 kW m ⁻² (mm)
REF	1.6 ± 0.3	2.4 ± 0.3
CH	2.0 ± 0.3	3.1 ± 0.4
CH-H	1.2 ± 0.1	2.6 ± 0.2
SC	2.0 ± 0.1	2.0 ± 0.3
SC-H	3.1 ± 0.7	1.8 ± 0.2
DC	1.1 ± 0.1	1.6 ± 0.2
DC-H	0.7 ± 0.1	1.0 ± 0.2

4. Discussions

The results of experimental evaluation of boiling heat transfer of functionalized surfaces have shown that all fabricated surfaces are capable of enhancing both the critical heat flux and the heat transfer coefficient in comparison with the untreated reference surfaces. Here, we aim to explain the mechanisms behind the enhancement and interpret the reasons for differences between different surface morphologies (i.e., the three types of surfaces—CH, SC and DC) and between the hydrophobic and hydrophilic variant of an individual surface (i.e., surfaces with or without the “-H” suffix).

Laser texturing modifies the surface morphology and surface chemistry of the treated surface. Typically, laser ablation as used within this study (and most others) causes the appearance of either singular craters or lines of craters, which form valleys surrounded by ridges of resolidified material. The latter is typically rich in oxide species and porous to